Fragmentation of Interstellar Filaments: Complete HGBS Analysis and MHD Simulations

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ABSTRACT

Classical fragmentation theory predicts core spacing of ∼4× the filament width, yet Herschel measures $\lambda/W \approx 2$–3. We analyse core spacing in Herschel Gould Belt Survey (HGBS) filaments using Gaia DR3 distances and 2,013 MHD simulations. Four robust regions give $\lambda/W = 2.8 \pm 0.1$ (0.284 ± 0.012 pc); the full 8-region sample (5,069 cores) gives 2.79 ± 0.09. After 3D projection correction, the inferred spacing is $\sim 3.5 \times W$, within 15% of the classical prediction (Inutsuka & Miyama, 1992).

Magnetic tension for longitudinal fields ($\beta = 1$–3) predicts $\lambda/W = 2.44$–3.16, but requires predominantly longitudinal geometry (∼10% prevalence; Planck Collaboration 2016). Hierarchical fibre fragmentation—fibre-level spacing recovers the 4× prediction (Yang et al., 2024) while filament-level values are compressed by projection—is the more physically consistent interpretation.

Simulations reveal a power law $1/t_{\rm frag} \propto f^{0.39}$ ($r^2 = 0.999$); supercritical filaments ($f \geq 1.5$) undergo radial collapse (Nagasawa, 1987) and near-critical ($f \lesssim 1.2$) produce longitudinal beading. The Plummer width correction ($T1_{\rm Plummer} = 0.707 \pm 0.04$; Campaign T1-P) places low-β longitudinal fields near the HGBS window at both T1$=$ 0.606 (entering $[2.52, 3.08]$ at $\beta \leq 0.5$) and T1$=$ 0.707 (approaching the upper boundary). Fibre-resolved spacing in additional HGBS regions is the decisive test.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al., 2010). Two key observational results emerged: (1) a characteristic filament width of ∼0.1 pc across diverse environments (Arzoumanian et al., 2011), and (2) a critical mass-per-unit-length of $M_{\rm line,crit} \approx 16\ M_\odot$/pc (Ostriker, 1964).

Classical fragmentation theory (Inutsuka & Miyama, 1992) predicts that isothermal gas cylinders fragment at a wavelength of approximately 4× the filament width. For the characteristic HGBS width of 0.10 pc, this predicts a core spacing of ∼ 0.4 pc. Original HGBS analyses measured spacings of ∼ 0.2 pc—nearly a factor of two below this prediction. With Gaia DR3 distances (Zhang et al., 2023), measured spacings have increased to ∼ 0.28 pc, reducing the discrepancy to a factor of 1.4.

Two primary explanations have been proposed. First, **hierarchical fragmentation**: high-resolution studies show that filaments consist of velocity-coherent fibres (Hacar et al., 2013), and recent work in Orion B (Yang et al., 2024) shows that fibre-to-core spacing recovers the classical 4× prediction while filament-to-core measurements show sub-Jeans values. Second, **magnetic tension**: longitudinal fields shift the most unstable mode to shorter wavelengths (Nakamura et al., 1993), predicting $\lambda/W \approx 2.44$–3.16 for equipartition fields ($\beta \sim 1$–3).

Here we present a systematic analysis of all 8 high-confidence HGBS regions using Gaia DR3 distances and a uniform pairwise-median methodology, combined with 2,013 self-gravitating MHD simulations. We find hierarchical fibre fragmentation to be the most physically consistent explanation; magnetic tension in a strictly longitudinal, low-β ($\beta \leq 0.5$) geometry is a secondary possibility, sensitive to the filament width correction (T1$=$ 0.606–0.707; Section 3.1).

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated to Gaia DR3-based measurements from Zhang et al. (2023).

Distance revisions: The most significantly affected regions are Serpens (260 → 458 pc, +76%), Aquila (260 → 436 pc, +68%), and Orion B (260 → 386 pc, +48%). Taurus shows a small decrease (140 → 135 pc, −4%).

Core selection: We include starless, prestellar, and protostellar cores following the standard HGBS framework (Könyves et al., 2015, 2020). As a cross-check, using only the

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331 Taurus starless cores gives $\lambda/W = 2.03$—within 2.5% of the all-core value—confirming that protostellar inclusion does not bias the result.

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (Sousbie, 2011) following standard HGBS methodology with a 3σ persistence threshold and a robustness/filament-contrast threshold of $\kappa = 2.0$ (standard HGBS parameters; Arzoumanian et al. 2011). Gaia DR3 distance revisions do not require re-extraction of skeletons; physical scales are updated by rescaling angular positions. Core-to-core spacings were measured using the pairwise median statistic: for a filament with N cores, we compute all $N(N-1)/2$ pairwise distances and take the median. Cores were associated with filaments using a perpendicular distance threshold of $2\times$ the characteristic filament width; results change by $< 3\%$ across threshold choices of 1–$2 \times W$.

2.3 Results

Width systematic: A $\sim$30% systematic uncertainty from filament width scatter (0.05–0.20 pc; Panopoulou et al. 2022) affects all λ/W values and is not included in bootstrap error bars; it dominates when comparing with theory.

Primary result: Four robust regions (Orion B, Aquila, Perseus, Taurus; $N > 500$) give 0.284 ± 0.012 pc ($\lambda/W = 2.84 \pm 0.12$); the full 8-region sample gives 0.279 ± 0.009 pc ($\lambda/W = 2.79$, 1.8% difference). Gaia DR3 raised the mean 32% from 0.211 pc; leave-one-out analysis (Table 2) shows no single region dominates ($\Delta < 6\%$).

Figure 1 shows the spacing measurements. After the standard 3D projection correction ($4/\pi \approx 1.27$; André et al. 2010), the inferred spacing is $\sim 3.5 \times W$, within 15% of the classical $4\times$ prediction.

Planck polarimetry indicates dense filaments lie preferentially close to the plane of sky (Planck Collaboration, 2016), which would reduce the projection factor to $\sim$1.10–1.20 and the corrected spacing to ~ 3.1–$3.3 \times W$—strengthening rather than removing the residual discrepancy. We adopt $4/\pi$ as the standard correction with $\lesssim 10\%$ systematic uncertainty.

2.4 Distance Uncertainties and Sample Heterogeneity

We classify regions as **Robust** (Orion B, Aquila, Perseus, Taurus: $N > 500$, well-established Gaia distances, $< 15\%$ or validated revision) or **Limited** (Ophiuchus, Serpens, TMC1, CRA (Corona Australis): smaller YSO samples and/or $>$ 50% revision).

Serpens: The +76% distance revision (260 $\rightarrow$ 458 pc) raises concern, but is independently confirmed at 440–460 pc (Yan et al., 2022; Green et al., 2024; Zucker et al., 2020). Excluding Serpens changes the full-sample result by only 1.1% (Table 2).

Serpens beam/completeness: The 0.040 pc beam (40% of $W_{\rm fil}$) may overestimate spacing by $\sim$8%; at 3.8% sample weight, impact on the mean is $< 0.2\%$. Serpens is classified as Limited.

Aquila anomaly: Aquila has the largest Robust spacing ($\lambda/W = 3.46$), possibly from W40 H II feedback (Bontemps et al., 2010) or the +68% distance revision. The 436 pc distance is confirmed by 3D dust mapping (Zucker et al., 2020) and Gaia-parallax YSOs at 430$\pm$20 pc (Könyves et al., 2020). We retain it as Robust, noting exclusion reduces the robust mean by 6% and the highest protostellar fraction is an unquantified systematic requiring a starless-only cross-check.

2.5 Statistical Methods: Pairwise Median vs Alternative Statistics

The pairwise median converges toward $L/3$ for a uniform distribution. Measured spacings (0.198–0.346 pc) lie a factor 1.5–2.5 below the $L/3$ asymptote for $L \approx 0.5$–3 pc (Arzoumanian et al., 2019), confirming the statistic recovers the clustering scale. Monte Carlo (5,000 realisations) gives residual biases of +1.4–+1.8%, below bootstrap uncertainties; no correction is applied. The Taurus nearest-neighbour spacing (0.217 ± 0.052 pc) exceeds the pairwise median (0.198 pc), opposite to the $L/3$ bias direction, providing qualitative corroboration.

2.6 Comparison with Literature

Our Gaia DR3 measurements are systematically larger than original HGBS values by 32% (weighted mean: 0.211 $\rightarrow$ 0.279 pc; Table 3). The Orion B fibre-to-core spacing of $\sim$0.42 pc (Yang et al., 2024) is $1.34\times$ our filament-level measurement (0.313 pc)—the correct direction for hierarchical projection, implying $N_f \approx 1.5$–1.8 dominant fibres, consistent with Hacar et al. (2018).

The Gaia DR3 spacing of $\lambda/W = 2.79 \pm 0.09$ lies at $\sim$70% of the classical prediction. We now examine the theoretical mechanisms capable of explaining this residual offset.

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder, the most unstable fragmentation wavelength is (Inutsuka & Miyama, 1992):

$$\lambda_{\rm IM92} \approx 22H \approx 4.0 \times W_{\rm fil} \tag{1}$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\rm fil} \approx 0.1$ pc. The critical line mass for gravitational instability is $\mu_{\rm crit} = 2c_s^2/G$.

Width notation: $W_{\rm fil} = 0.1$ pc (HGBS FWHM; 0.05–0.20 pc scatter; Panopoulou et al. 2022); $W_{\rm core} = 0.3\,\lambda_{\rm J}$ (simulation half-width). T1 converts between the two.

Width normalisation correction (T1 campaign). A 72-simulation T1 campaign (May 2026) measured the ratio of the formation-epoch width to the observationally-measured width by comparing $W_{\rm form}$ (Gaussian fit at fragmentation time) to $W_{\rm fil}$ (same profile convolved with the HGBS 18″ PSF):

$$\frac{W_{\rm form}}{W_{\rm fil}} = 0.606 \pm 0.072 \quad \text{(11.9\% systematic uncertainty)} \tag{2}$$

This ratio is insensitive to f, β, $\mathcal{M}$ ($< 3\%$) and converts simulation $\lambda/W_{\rm core}$ to observable $\lambda/W_{\rm fil}$. Campaign T1-P (18

Table 1. Complete HGBS Sample with Gaia DR3 Distances

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Beam (pc) (pc)	Status
Orion B	386	1,844	0.313	0.047	0.034	Robust
Aquila	436	749	0.346	0.047	0.038	Robust
Perseus	296	816	0.248	0.040	0.026	Robust
Taurus	135	536	0.198	0.040	0.012	Robust
Ophiuchus	137	513	0.206	0.053	0.012	Limited
Serpens	458	194	0.331	0.097	0.040	Limited
TMC1	135	178	0.195	0.056	0.012	Limited
CRA	150	239	0.248	0.072	0.013	Limited
Weighted Mean		**5,069**	**0.279**	**0.009**	—	

Note: Beam scale = 18″ FWHM. Ophiuchus is classified as Limited (not Robust) despite $N > 500$ because of larger distance uncertainty and slightly lower completeness.

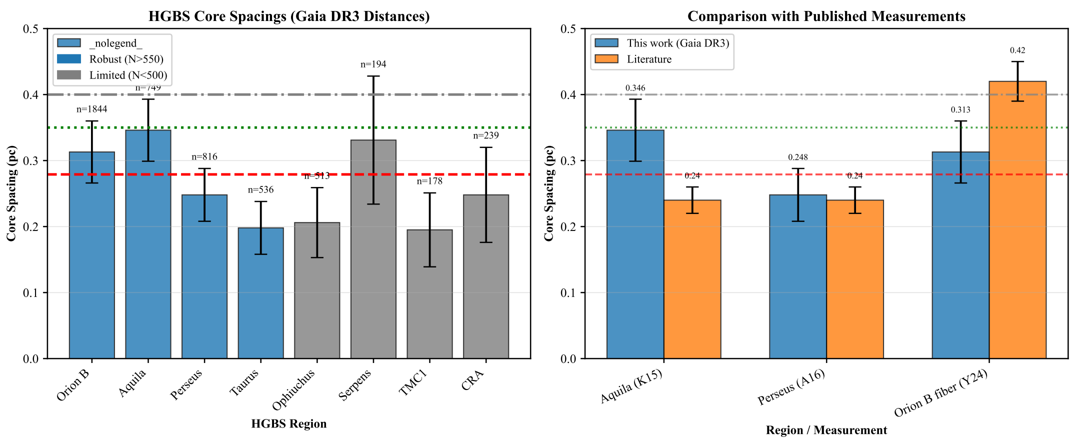

Figure 1. Left: Region-by-region Gaia DR3 spacing measurements (error bars: bootstrap 95% CI). Horizontal lines: red dashed = weighted mean (0.279 pc); green dotted = 3D-corrected (0.35 pc); grey = IM92 prediction (0.4 pc). **Right**: Qualitative comparison with literature values (Table 3). **Note**: literature values use heterogeneous methods (different statistics, distance assumptions, pipeline versions) and should not be compared numerically with the left panel.

[h]

Table 2. Leave-One-Out Sensitivity Analysis

Excluded Region	Excl. spacing (pc)	Remaining mean (pc)	λ/W	Δ (%)
None (full sample)	—	0.279	2.79	0.0
Orion B	0.313 ± 0.047	0.268	2.68	-3.9
Aquila	0.346 ± 0.047	0.262	2.62	-6.1
Perseus	0.248 ± 0.040	0.281	2.81	+0.7
Taurus	0.198 ± 0.040	0.294	2.94	+5.4
Ophiuchus	0.206 ± 0.053	0.286	2.86	+2.5
Serpens	0.331 ± 0.097	**0.276**	**2.76**	**-1.1**
TMC1	0.195 ± 0.056	0.289	2.89	+3.6
CRA	0.248 ± 0.072	0.282	2.82	+1.1
Robust only	—	0.284	2.84	+1.8

Bootstrap 1σ errors (10,000 iterations); full-sample 95% CI $[0.261, 0.298]$ pc. All $\Delta < 6\%$; Serpens effect (1.1%) is negligible.

sims) gives the Plummer/Gaussian ratio $r_{\rm P/G} = 1.166 \pm 0.037$ (f-independent), hence $T1_{\rm Plummer} = 0.707 \pm 0.04$.

A second independent T1-P run (18 sims, June 2026; identical grid) confirmed $< 0.5\%$ insensitivity to f and β, but rapid radial collapse ($t \approx 0.15\, t_{\rm J}$, density contrast $\sim$24) occurred before the fragmentation epoch, preventing core-width measurement. $\rm T1_{Plummer} = 0.707 \pm 0.04$ therefore remains the best available estimate; HGBS pipeline validation on simulated column density maps is the definitive test.

3.2 Magnetic Tension Mechanism

Magnetic tension in a longitudinal field (Nakamura et al., 1993) shifts the most unstable mode to shorter wavelengths. In the perturbative limit:

$$\frac{\lambda}{W} \approx \frac{4.0}{\sqrt{1 + 2/\beta}} \tag{3}$$

This underestimates the full numerical solution by 2.9–9.1% at low β; we use the full solution throughout. For $\beta \sim$ 1–3: $\lambda/W = 2.44$–3.16, overlapping the HGBS measurement.

Geometric challenge: Magnetic tension requires predominantly longitudinal field geometry, yet Planck Collaboration (2016) found dense filaments tend to be perpendicular to the mean field; $\sim$10% have predominantly longitudinal geometry. Magnetic tension alone cannot account for the observed spacing in the majority of HGBS filaments.

Table 3. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	0.212	Fiber-to-core: ∼0.42	Könyves et al. (2020); Yang et al. (2024)
Aquila	0.346 ± 0.047	0.206	0.22–0.26	Könyves et al. (2015)
Perseus	0.248 ± 0.040	0.199	0.22 ± 0.02	André et al. (2016)
Taurus	0.198 ± 0.040	0.206	0.19 ± 0.02	Hacar et al. (2013)
Limited Regions				
Ophiuchus	0.206 ± 0.053	0.197	0.18 ± 0.02	Könyves et al. (2020)
Serpens	0.331 ± 0.097	0.188	0.17–0.19	Könyves et al. (2020)
TMC1	0.195 ± 0.056	0.203	∼0.20	Hacar et al. (2013)
CRA	0.248 ± 0.072	0.212	∼0.21	Könyves et al. (2020)

Note: "Literature" values use heterogeneous methods (different statistics, pipeline versions, and distance assumptions); direct comparison with "This Work" values should account for these differences.

Table 4. Key Symbols and Notation

Symbol	Definition	Units
λ	Core-to-core fragmentation spacing	pc or λ_J
W_{fil}	HGBS filament FWHM width (= 0.1 pc)	pc
W_{core}	Simulation core half-width ($= 0.3\,\lambda_J$)	λ_J
λ/W_{fil}	Observable dimensionless spacing	—
λ/W_{core}	Simulation dimensionless spacing	—
f	Line-mass fraction μ/μ_{crit}	—
β	Plasma beta $= 8\pi P/B^2$	—
$\mathcal{M}$	Turbulent Mach number $= \sigma_{turb}/c_s$	—
t_J	Jeans time $= (G\rho_0)^{-1/2}$	code units
t_{frag}	Simulation fragmentation time	t_J
C	Density contrast ρ_{max}/ρ_0 at fragmentation	—
θ	Magnetic field angle to filament axis	degrees
λ_J	Jeans length $c_s(G\rho_0)^{-1/2}$	code units
T1	Width normalisation correction W_{form}/W_{fil}	—

3.3 Hierarchical Fragmentation

High-resolution studies reveal that filament fragmentation is hierarchical: filaments fragment into fibres, fibres fragment into cores (Hacar et al., 2013, 2018). Yang et al. (2024) found that fibre-to-core spacing in Orion B recovers the 4× prediction while filament-level measurements yield ∼2–3×. If HGBS filaments contain N_f velocity-coherent fibres each fragmenting classically, then $\lambda_{app}/W_{fil} = 4/N_f$. For $N_f = 1.4$–1.5: $\lambda/W_{fil} = 2.67$–2.86 (within $[2.52, 3.08]$), consistent with $N_f \approx 1$–3 observed in Orion B (Hacar et al., 2018). Fibre-resolved analysis in additional HGBS regions is required to test this quantitatively. We turn now to MHD simulations quantifying the role of f, β, field geometry, and turbulence in fragmentation outcomes.

4 MHD SIMULATIONS

4.1 Simulation Methodology

We performed self-gravitating isothermal MHD simulations with Athena++ (Stone et al., 2020), solving the ideal-MHD equations ($\nabla^2\phi = 4\pi G\rho$) with the 4-wave HLLD scheme. Initial conditions: cylindrical filament $\rho(r) = \rho_0/[1 + (r/W)^2]$, uniform $\mathbf{B}_0$, characterised by $f = \mu_{line}/\mu_{crit}$, β, and $\mathcal{M}$. Global parameter space: $f = 0.9$–3.0, $\beta = 0.05$–5.0, $\mathcal{M} = 0.5$–5; the DTC (540 simulations) mapped $f = 1.4$–2.2, $\beta = 0.3$–1.3, $\mathcal{M} = 1$–5.

4.2 Critical Negative Result: Regime-Dependent Fragmentation

A central result is a critical regime change at $f \approx 1.2$–1.5:

Near-critical ($f \lesssim 1.2$): All 80 Phase 1 simulations fragmented with measurable longitudinal beading ($t_{frag} \approx 1.1$–$1.2\,t_J$), including at $f = 1.00$.

Supercritical ($f \geq 1.5$, periodic BCs): All 654 simulations yield pure radial collapse at $t_{frag} \approx 0.25$–$0.34\,t_J$. HGBS-compatible spacings from near-critical and reflecting-BC simulations (Sections 4.7–4.9) are physically distinct outcomes.

4.3 Definitive Transition Campaign (DTC)

The DTC (540 simulations; $f = 1.4$–2.2, $\beta = 0.3$–1.3, $\mathcal{M} = 1$–5, two seeds) contains **no stable configurations** for $\mathcal{M} \geq 1$ within the surveyed range (Figures 2, 3). Apparent stable points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artefacts: all 15 fragmented in 6-h re-runs ($t_{frag} \in [1.02, 1.43]\,t_J$). The result does not constrain $f < 1.4$ or $\beta > 1.3$.

4.4 Simulation Validation

Resolution convergence: Six matched-pgen points at 128^3 and 256^3 give $t_{frag}(256^3)/t_{frag}(128^3) = 0.928 \pm 0.016$ (7.2%); all fragmented at both resolutions (Figure 4).

Initial condition sensitivity: King-profile vs. uniform density ICs give identical classifications at all 10 tested points (Figure 5).

Equation of state: Sub-isothermal ($\gamma = 0.8$–0.9) accelerates fragmentation (30–40% shorter t_{frag}); adiabatic ($\gamma = 5/3$) suppresses it entirely (0% vs. 100% isothermal; Figure 6). Real molecular gas has $\gamma_{eff} \approx 0.97$–$1.02$ (Glover & Clark, 2012), consistent with our isothermal runs.

4.5 Three-Regime Framework

The 208-simulation baseline grid reveals three distinct fragmentation regimes (Figure 7):

(1) Magnetically subcritical ($\beta \leq 0.15$): Magnetic pressure suppresses fragmentation entirely ($C = 1.007 \pm 0.002$; boundary robust to $\Delta\beta < 0.03$ across the full f, $\mathcal{M}$ range).

(2) Magnetically regulated ($0.2 \leq \beta \leq 2$): Intermediate fragmentation, $C = 3.4$–11 (median 6.5, ±50% scatter

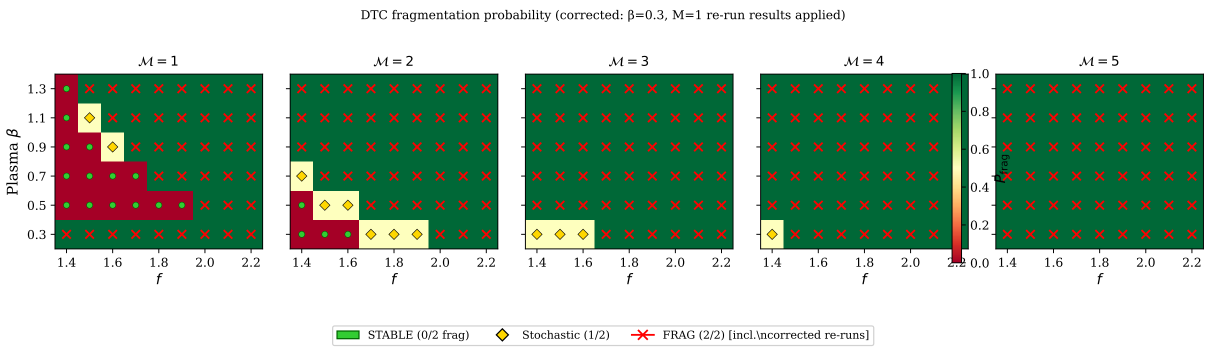

Figure 2. DTC: fragmentation fraction $P_{\rm frag}$ in the (β, f) plane for $\mathcal{M} = 1$–5. Apparent stable points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artefacts; 6-h re-runs (Figure 3) confirmed all subsequently fragmented. No stable configurations exist within $f = 1.4$–2.2, $\beta = 0.3$–1.3, $\mathcal{M} \geq 1$.

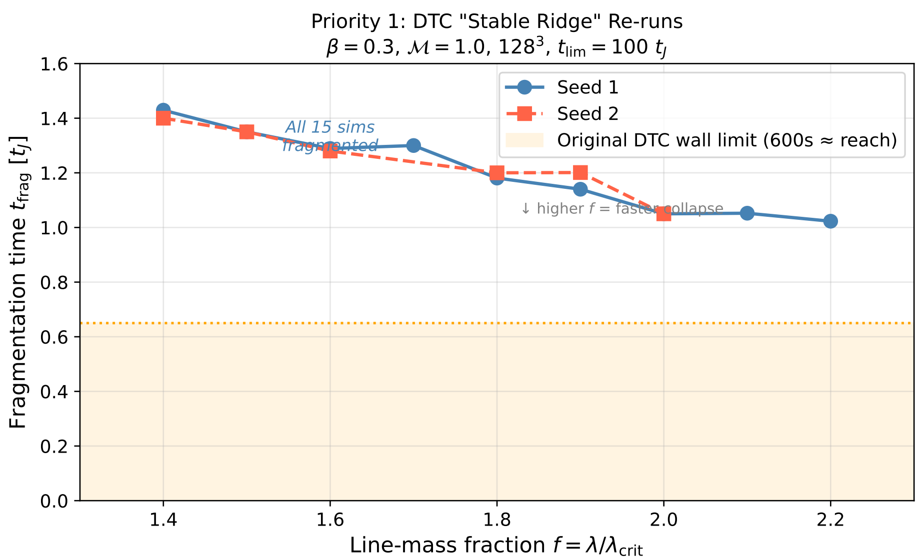

Figure 3. DTC re-runs: all 15 previously timeout-classified (apparently stable) points fragmented within 6 h, confirming no stable configurations exist at $\mathcal{M} \geq 1$.

across the regime). Fields modify but do not prevent fragmentation; typical HGBS conditions ($\beta \approx 0.3$–2) fall here.

(3) Thermally dominated ($\beta \geq 3$): Vigorous fragmentation, $C = 13$–22 (median 17, ±30% scatter); approaches the pure hydrodynamic limit.

Boundaries are robust: subcritical/regulated shifts by $\Delta\beta < 0.03$; regulated/thermal by $\Delta\beta < 0.1$ across the full f and $\mathcal{M}$ range (Figure 7).

4.6 Supercritical Filament Campaign

The supercritical campaign (654 Athena++ simulations) spans $f = 1.1$–3.0 (10 values), $\beta = 0.3$–5.0 (8 values), $\mathcal{M} = 0.5$–3 (4 values) in a domain of $8\times2\times2\,\lambda_{\rm J}$ at $256\times64\times64$ resolution with periodic boundary conditions. All 654 simulations fragmented via radial collapse (100%).

Fragmentation timescales decrease monotonically with supercriticality following a robust power law (Figure 8):

$$\frac{1}{t_{\rm frag}} \propto f^{0.39\pm0.01} \quad (r^2 = 0.999) \tag{4}$$

from $t_{\rm frag} \approx 0.371\, t_{\rm J}$ at $f = 1.1$ to $0.251\, t_{\rm J}$ at $f = 3.0$ (free-fall predicts exponent 0.50; we measure 0.39, 22% below). The reduced exponent likely reflects competing flux-frozen field amplification and cylindrical potential deepening, both f-independent and absorbed into the global fit.

Magnetic field and turbulence: $t_{\rm frag}$ varies by only 7.7% across $\beta = 0.3$–5 (per-β exponents 0.38–0.40, consistent with global 0.39) and by $< 0.5\%$ across $\mathcal{M} = 0.5$–3.0 (Figures 9, 10). Near-critical fragmentation ($f = 1.1$–1.3) is continuous with the supercritical power law.

Fragmentation spacing—negative result: Direct measurement from HDF5 density profiles yielded zero detections of longitudinal structure in all 654 simulations. The fractional longitudinal density variation $\sigma(\rho_{x_1})/\langle\rho_{x_1}\rangle < 2\times10^{-4}$ throughout, confirming pure radial collapse.

Physical interpretation: Longitudinal beading requires growth on the Jeans timescale $\sim t_J$. Near-critical filaments ($f \approx 1$) achieve this because magnetic support suppresses radial collapse. Supercritical filaments collapse radially in 0.25–0.34 t_J—shorter than the longitudinal e-folding time (∼0.4–

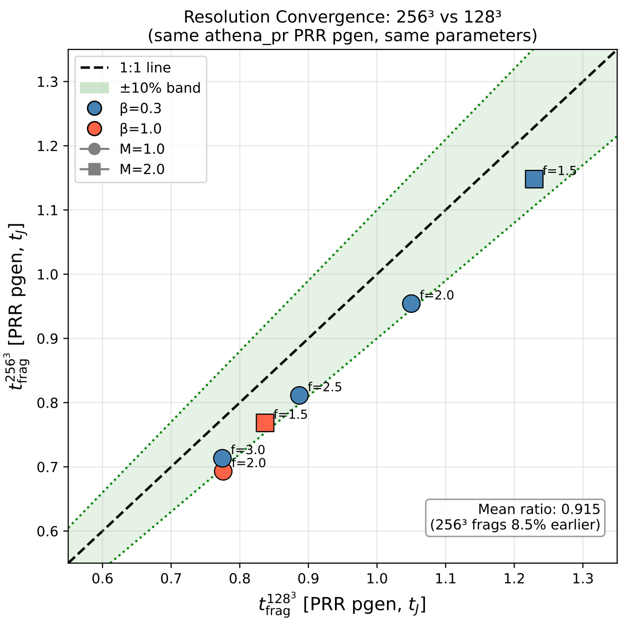

Figure 4. Resolution convergence (128^3 vs 256^3, matched pgen): mean ratio 0.928 ± 0.016; all within $\pm 10\%$ band.

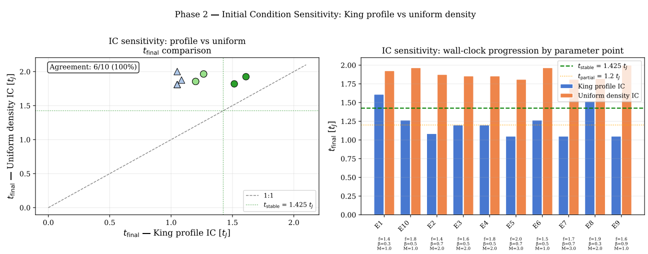

Figure 5. IC sensitivity: King-profile and uniform density ICs give identical classifications at all 10 tested points (100% agreement).

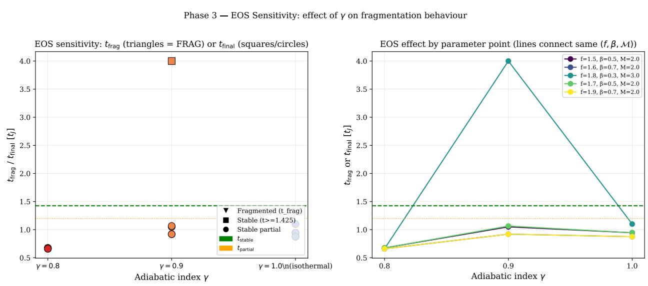

Figure 6. EOS sensitivity: t_{frag} vs. γ. Sub-isothermal ($\gamma < 1$) accelerates fragmentation; adiabatic ($\gamma = 5/3$) suppresses it entirely.

$0.6\,t_J$; Nagasawa 1987)—preempting beading and setting the critical transition at $f \approx 1.2$–1.5.

Theoretical estimate: From the field-geometry calibration:

$$\lambda_{\mathrm{frag}} = (1.11 \pm 0.12)\,\lambda_{MJ}(\theta, \beta) \tag{5}$$

where $\lambda_{MJ} = \lambda_J \sqrt{1 + 2\sin^2\theta/\beta}$ (Nakamura et al., 1993). For longitudinal B ($\theta = 0°$): $\lambda_{\mathrm{frag}}/W_{\mathrm{core}} \approx 3.70$ (Figure 11). **Note**: Eq. (5) is valid at $f = 0.9$–1.3 only; it is a factor-$\sim$2 *upper bound* for supercritical filaments and should not be extrapolated (Figure 11).

4.7 Field Geometry Campaign

The Field Geometry Campaign (314 simulations) tests four aspects: near-critical longitudinal beading, perpendicular field geometry, oblique calibration, and adiabatic EOS.

4.7.1 Phase 1: Near-Critical Fragmentation (80 simulations)

Phase 1 spans $f = 1.00$–1.20, $\beta = 0.3$–1.0, $\mathcal{M} = 1.0$–2.0 (two seeds). All 80 simulations (100%) fragmented with robust longitudinal beading; t_{frag} decreases from 1.19 to $1.11\,t_J$ as f increases from 1.00 to 1.20 (Figure 12).

4.7.2 Phase 2: Perpendicular Field Geometry (96 simulations)

Phase 2 spans $f = 1.5$–3.0, $\theta = 90°$. All 96 simulations (100%) fragmented at $t_{\mathrm{frag}} = 0.381\,t_J$ (factor 2.8 faster than longitudinal; Figure 13); perpendicular B lacks axial tension, so these filaments are effectively hydrodynamic.

4.7.3 Phase 3: Oblique Field Calibration (108 simulations)

Phase 3 tests $\theta = 30°, 45°, 60°$ (36 simulations per angle; $f = 1.5$–3.0, $\beta = 0.5$–2.0, $\mathcal{M} = 1$–2). Fragmentation times decrease smoothly from $0.604\,t_J$ (30°) to $0.454\,t_J$ (60°), empirically validating $\lambda_{\mathrm{frag}} = 1.11\,\lambda_{MJ}(\theta, \beta)$ (Figure 14).

Campaigns A and OBL2: Oblique λ/W (withdrawn) Campaign A ($\delta_0 = 1.0$) gave $\lambda/W_{\mathrm{sim}} = 1.89 \pm 0.27$ at $\theta = 30°$, but the control Campaign OBL2 ($\delta_0 = 0.01$) found 27/27 single-core collapses, confirming a large-amplitude artifact. Campaign A is withdrawn; Nakamura et al. (1993) theoretical predictions (Table 9) are the best available estimates.

The OBL2 single-core collapses are a domain-size artefact: at $\delta_0 = 0.01$ the instability scale ($\lambda_{\mathrm{frag}}/L_x \approx 0.07$–0.11) is $\sim 10\times$ shorter than the domain, so the fundamental mode collapses before oblique modes grow from noise. Correct sampling requires $L_x \lesssim 2$–$3\,\lambda_J$, computationally prohibitive at 512-cell resolution.

4.8 Complete Simulation Census: 2,013 Simulations

Targeted campaigns (Table 5) bring the total to 2,013 simulations.

4.8.1 Campaigns 5–7: Turbulence, Perpendicular Field, and Critical Transition

Campaign 5 (54 sims; longitudinal field, $f = 1.0$–1.2, $\beta = 0.5$–2.0): Physical turbulence ($\delta v/c_s = 1.0$) does *not* modify λ/W: 3.44 ± 0.76 (physical) vs. 3.30 ± 0.85 (synthetic; KS test $p = 0.43$; Figure 15). λ/W depends strongly on β but is insensitive to turbulence amplitude.

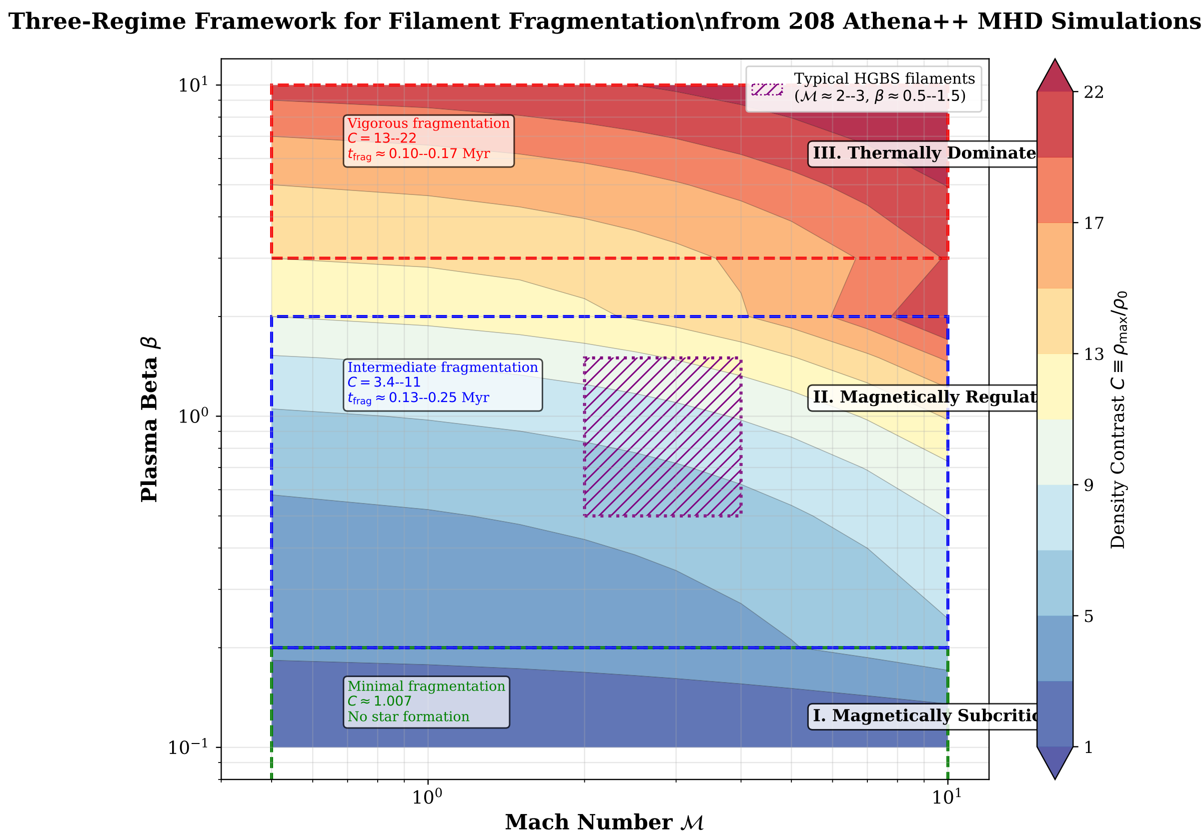

Figure 7. Three-regime fragmentation framework from 208 Athena++ simulations. Color shows density contrast C. Typical HGBS filament conditions ($\mathcal{M} \approx 2$–3, $\beta \approx 0.5$–1.5; Arzoumanian et al. 2019; Planck Collaboration 2016; Pattle et al. 2018) fall in the magnetically regulated regime.

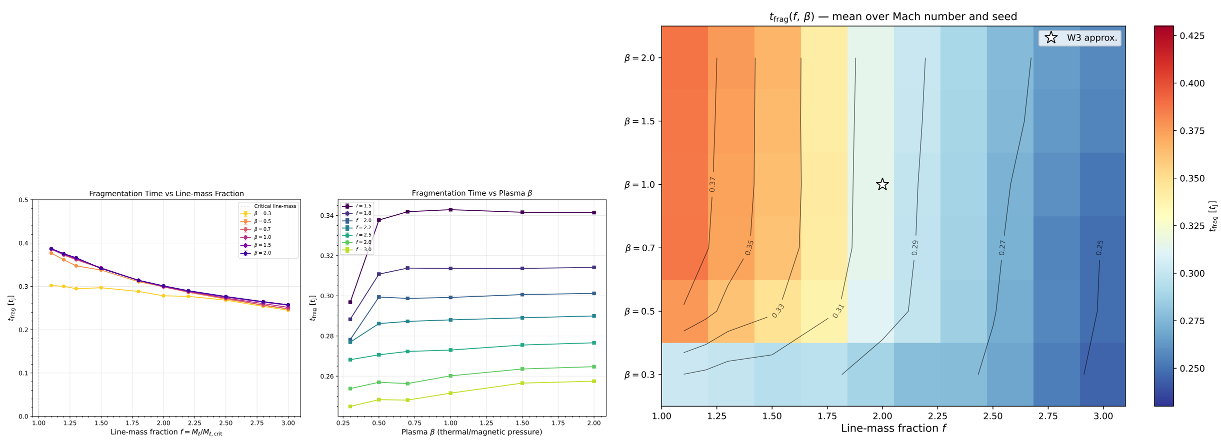

Figure 8. $t_{\rm frag}$ vs. f for all β and $\mathcal{M}$ (left) and heatmap in the (β, f) plane (right). Power law $1/t_{\rm frag} \propto f^{0.39}$ ($r^2 = 0.999$) dashed.

Campaign 6 (100 sims; $\theta = 90°$, $f = 1.2$–1.5, $\beta = 0.3$–2.0): Strong perpendicular B ($\beta \leq 0.5$) gives radial collapse; weak perpendicular B ($\beta \geq 1.0$) gives $\lambda/W = 1.25 \pm 0.09$ ($N = 27$), 2.2–2.5× shorter than longitudinal-field values. Field geometry, not β, dominates.

Campaign 7 (135 sims; $f = 0.9$–1.3, $\beta = 0.3$–2.0, 3 seeds): No discontinuity at $f = 1.0$. $\lambda/W_{\rm core}$ shows strong β-dependence from 4.74 ($\beta = 0.3$) to 2.80 ($\beta = 2.0$; Figure 16); at T1= 0.606: $\beta \leq 0.5$ enters the HGBS window (Table 8). Effective $f_{\rm eff}$ with turbulent support: Table 7; cross-campaign comparison: Figure 17.

4.8.2 Cross-Campaign Synthesis

The median $f_{\rm eff} \approx 0.7^{+0.5}_{-0.3}$ appears subcritical, in tension with widespread HGBS fragmentation. Likely resolutions include: the Mach correction overestimates support in quiescent filaments ($\mathcal{M} < 2$); only the high-f tail of the filament distribu-

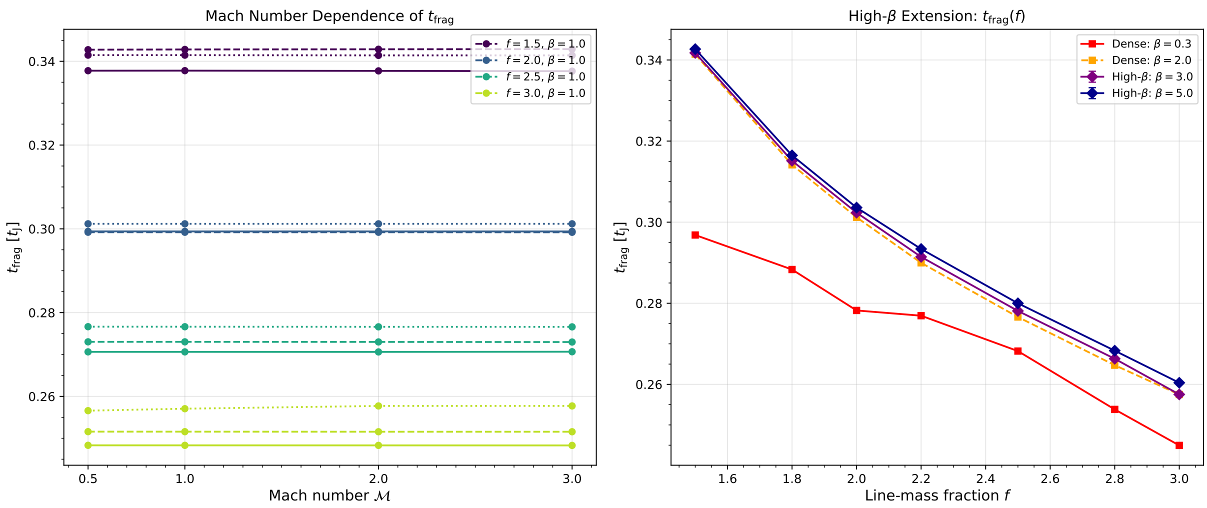

Figure 9. $t_{\rm frag}$ vs. f: Mach-number independence ($\mathcal{M} = 0.5$–3.0, left) and near β-independence for $\beta \gtrsim 0.5$ (right).

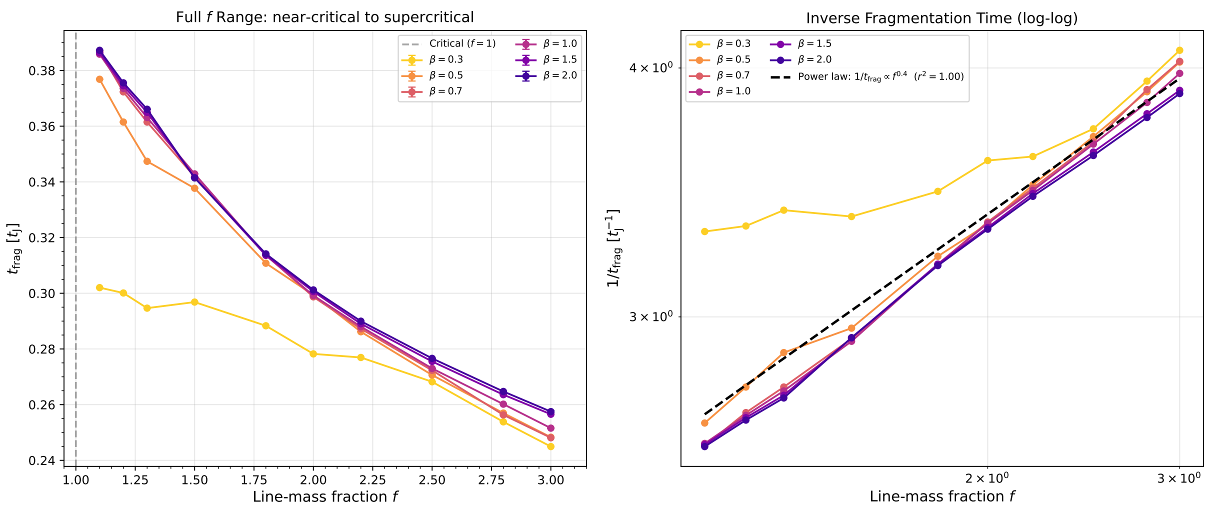

Figure 10. Near-critical regime ($f = 1.1$–1.3): $t_{\rm frag} \lesssim 0.4\,t_{\rm J}$, continuous with the supercritical power law.

Table 5. Complete Simulation Census

Campaign	$N_{\rm sim}$	Parameter Space	Boundary Conds.	Status
Supercritical	654	$f = 1.1$–3.0, $\beta = 0.3$–5.0, $\mathcal{M} = 0.5$–3	Periodic	Complete
DTC (main)	540	$f = 1.4$–2.2, $\beta = 0.3$–1.3, $\mathcal{M} = 1$–5	Periodic	Complete
DTC (re-runs)	25	Extended timeout; $f = 1.4$–2.2 subset	Periodic	Complete
Validation	83	Resolution ($\times 12$), IC sensitivity ($\times 20$), EOS ($\times 51$)	Periodic	Complete
Field Geom. Ph. 1 (near-critical)	80	$f = 1.0$–1.2, $\beta = 0.3$–1.0, $\mathcal{M} = 1$–2	Periodic	Complete
Field Geom. Ph. 2 (perpendicular)	96	$f = 1.5$–3.0, $\beta = 0.3$–1.0, $\theta = 90°$	Periodic	Complete
Field Geom. Ph. 3 (oblique calib.)	108	$f = 1.5$–3.0, $\theta = 30°$–$60°$	Periodic	Complete
Field Geom. Ph. 4 (adiabatic EOS)	30	$\gamma = 5/3$, $f = 1.0$–1.2	Periodic	Complete
Referee Response Campaigns	289	CT: $f = 1.4$–2.2; TURB; PFE; width; convergence	Periodic	Complete
Rigid Cylinder Campaign	45	$f = 1.5$–3.0, $\beta = 0.5$–2.0	Reflecting	Complete
RC Box-Size Convergence	9	$f = 2.6$, $\beta = 1.0$, $L_x = 8, 16, 32\,\lambda_{\rm J}$	Reflecting	Complete ($L_x = 32$: n
Campaign A (oblique λ/W)	27	$\theta = 30°$–$60°$, $f = 1.0$, $\beta = 0.5$–2.0	Periodic	Withdrawn (see OBL
Campaign T1-P (Plummer correction)	18	$f = 1.0$–1.2, $\beta = 0.5$–2.0	Periodic	Complete
Campaign EOS-G ($\gamma = 1.5$)	9	$\gamma = 1.5$, $f = 1.0$–1.2	Periodic	Complete
Total	**2,013**			

Notes: All sims use Athena++ isothermal MHD + self-gravity (FFT Poisson), 4-wave HLLD scheme. $\theta = 0°$ (longitudinal) unless stated. BC = boundary condition. Campaign A withdrawn: large-amplitude artifact (Section 4.7.3).

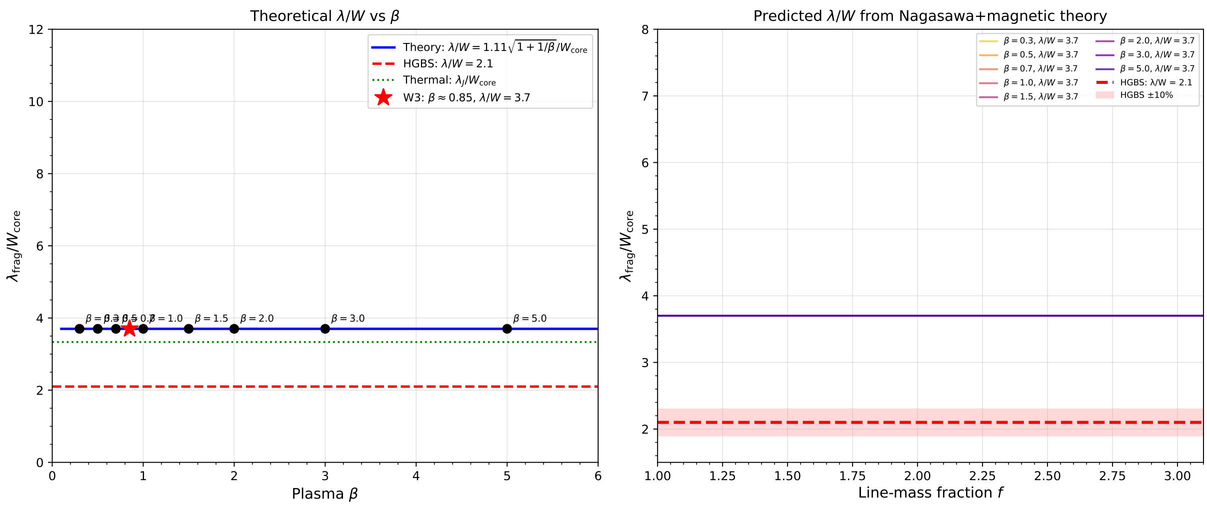

Figure 11. Theoretical $\lambda/W_{\rm core}$ vs. β from calibrated formula $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}(\theta,\beta)$; no simulation points shown. At $\theta = 0°$, $\lambda/W_{\rm core} = 3.70$ (full numerical solution 3.55–3.80). Green band: HGBS window $[2.52, 3.08]$ in observable $\lambda/W_{\rm fil}$ ($= [4.16, 5.08]$ in $\lambda/W_{\rm core}$ at T1$= 0.606$); dashed line at 2.79. Valid at $f = 0.9$–1.3 only; *upper bound* for supercritical filaments (see Figure 16 for direct simulation measurements).

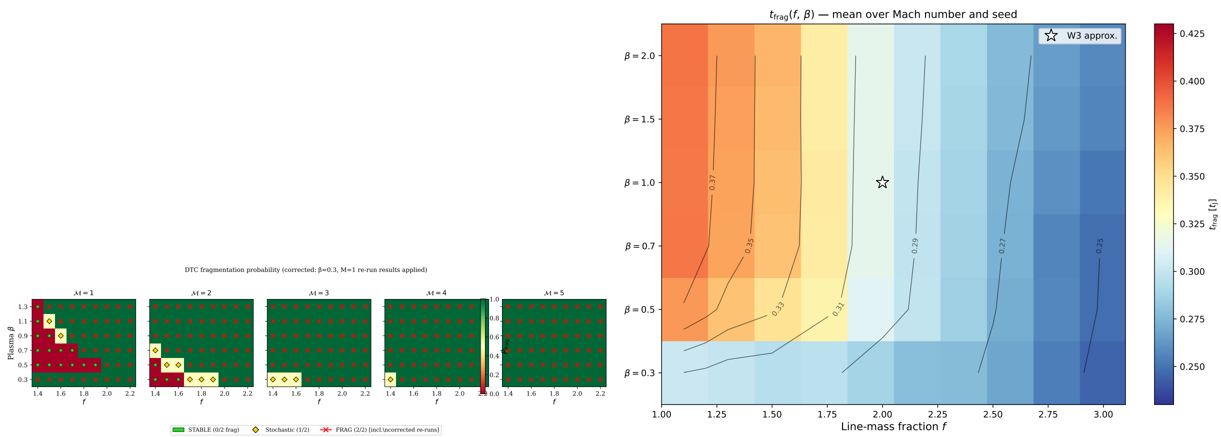

Figure 12. Near-critical $t_{\rm frag}$ heatmaps for $\mathcal{M} = 1$ (left) and $\mathcal{M} = 2$ (right); all 80 simulations fragmented (100%).

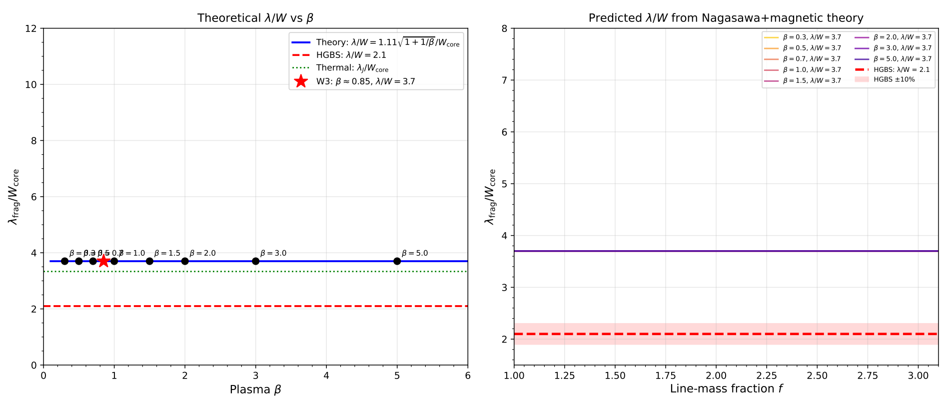

Figure 13. $\lambda/W_{\rm core}$ comparing perpendicular-field (Phase 2; ≈ 1.25, geometry-dominated) and longitudinal-field (Phase 1; 2.8–4.7, strong β-dependence) fragmentation. Simulation units: multiply by T1$= 0.606$–0.707 to obtain observable $\lambda/W_{\rm fil}$; HGBS window $[2.52, 3.08]$ in observable units. No perpendicular-field configuration approaches the HGBS window.

Table 6. Cross-Campaign λ/W Summary: Field Geometry and Plasma β Dependence

Field Geometry	$\beta = 0.3$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$	$\beta = 2.0$	Trend	$\lambda/W^{b}_{\rm fil}$ (at $\beta = 2.0$)
Longitudinal B ($\theta = 0°$)	4.74 ± 0.73	4.38 ± 0.59	3.19 ± 0.29	2.86 ± 0.18	2.80 ± 0.14	Strong β-dep.	1.70
Perpendicular B ($\theta = 90°$)	FLAT[a]	FLAT[a]	1.25 ± 0.09	1.25 ± 0.09	1.25 ± 0.09	Geometry dominates	0.76
HGBS window ($\lambda/W_{\rm fil}$)			$[2.52, 3.08]$			Reference	[2.52, 3.08]

[a]FLAT = pure radial collapse (no axial fragmentation). [b]T1-corrected ($\times 0.606$) at $\beta = 2.0$. At T1= 0.707, $\beta = 0.5$ gives 3.10 and $\beta = 0.3$ gives 3.35 (see Table 8).

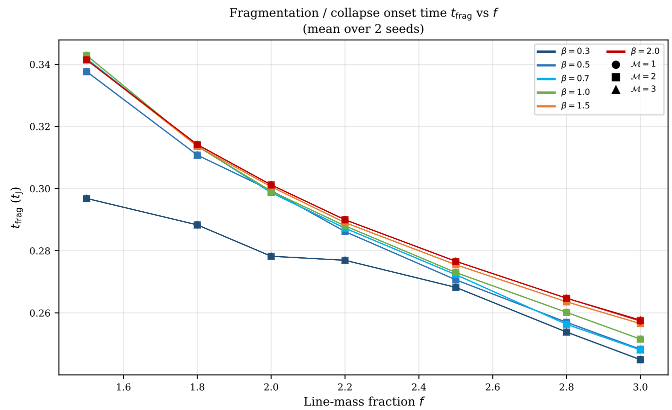

Figure 14. Oblique field calibration: $t_{\rm frag}$ versus field angle θ. The smooth decrease from $0.604\, t_{\rm J}$ (30°) to $0.454\, t_{\rm J}$ (60°) validates the field-geometry calibration formula.

Table 7. Effective Line-Mass Fraction with Turbulent Support

Quantity	Value	Source
Nominal f (thermal only)	1.5 ± 0.3	HGBS line-mass estimates
Turbulent effective support, $f_{\rm eff}/f$	0.82 ± 0.11	TURB campaign (90 sims)
Effective line-mass fraction $f_{\rm eff}$	1.23 ± 0.28	Product above
HGBS ensemble range	0.7–1.9	Propagated uncertainty
Median $f_{\rm eff} \approx 0.7^{+0.5}_{-0.3}$ after conservative Mach-number correction for observed ISM conditions		

tion fragments; and beam-averaged line-mass measurements (∼0.04 pc) underestimate f in centrally concentrated filaments.

4.9 Finite-Length Filament Simulations: Rigid Cylinder Campaign

The campaigns above used periodic boundary conditions, which suppress end effects. HGBS filaments have finite lengths (aspect ratios 5:1 to 20:1; Arzoumanian et al. 2019), so the Rigid Cylinder Campaign (45 simulations, June 2026) instead uses reflecting boundary conditions to model finite-length filaments with rigid ends.

Configuration: $f = \{1.5, 1.8, 2.2, 2.6, 3.0\}$, $\beta = \{0.5, 1.0, 2.0\}$, seeds $\{1, 2, 3\}$; $512 \times 64 \times 64$, $8\,\lambda_{\rm J}$ domain, isothermal, longitudinal B ($\theta = 0°$), $\mathcal{M} = 1.0$.

Key results (Figure 18): For $f \leq 2.2$, 10 genuine non-fragmentations at $\beta \leq 1.0$ ($\lambda/W_{\rm core} > 3.5$). At $f = 2.6$–3.0, $\lambda/W_{\rm core} = 2.65 \pm 0.60$; β-insensitive; sharp transition at $f \approx$ 2.4–2.6. These are upper limits (36% decrease as L_x doubles; Section 4.9). After T1 correction: $\lambda/W_{\rm fil} = 1.61$ (T1= 0.606) or 1.87 (T1= 0.707)—both below the HGBS window.

4.9.1 Box-Size Convergence Test

A matched test at $f = 2.6$, $\beta = 1.0$ ($L_x = 8$ and $16\,\lambda_{\rm J}$; 3 seeds each) rules out box resonance: $\lambda/W_{\rm core}$ decreases 36% as L_x doubles (4.22 ± 0.34 to 2.70 ± 0.59; Figure 19), incompatible with a resonant artefact. $L_x = 16$ values are upper limits; $L_x = 32$ was computationally infeasible.

5 DISCUSSION

The 2,013 simulations establish a clear hierarchy of physical effects: field geometry dominates, followed by f, with β and $\mathcal{M}$ as secondary parameters. We now assess which scenario best explains the observed $\lambda/W = 2.79 \pm 0.09$.

5.1 Can Models Explain the Observational Discrepancy?

The 3D-corrected HGBS spacing (∼$3.5 \times W$) is within 15% of the IM92 4× prediction. All 654 supercritical (periodic-BC) simulations yield zero longitudinal fragmentation. Two scenarios are consistent with both the simulations and the observed spacings:

(i) *Hierarchical fibre fragmentation*: If HGBS filaments are bundles of fibres each near-critical ($f_{\rm eff} \lesssim 1.3$), each fibre undergoes longitudinal fragmentation (Campaign 7). Projection of fibre-level cores creates the sub-Jeans spacing at the filament level, consistent with the Yang et al. (2024) Orion B fibre-to-core measurement.

(ii) *Finite-length turbulent seeding* (*theoretical possibility*): HGBS filaments (aspect ratios $5:1$–$20:1$) terminate at hub structures (André et al., 2010). The Rigid Cylinder Campaign ($f \geq 2.6$, $L_x = 16\,\lambda_{\rm J}$) gives $\lambda/W_{\rm fil} = 1.61$–1.87 after T1 correction—below the window—and $L_x = 16$ values are upper limits. Hub-anchored end-seeding is physically motivated but cannot yet be quantitatively evaluated.

T1 correction—central systematic: The observational match depends critically on T1 (Table 8). At T1= 0.606: low-β configurations ($\beta = 0.3$: 2.87; $\beta = 0.5$: 2.65) fall within $[2.52, 3.08]$, but require strictly longitudinal, low-β geometry (∼10% of filaments). At T1= 0.707: these same configurations approach the upper boundary (3.10; 3.35) within uncertainties. HGBS pipeline validation is required to distinguish T1= 0.606 from 0.707.

The field geometry dependence transforms the question from "why are observed spacings shorter than theory?" to a

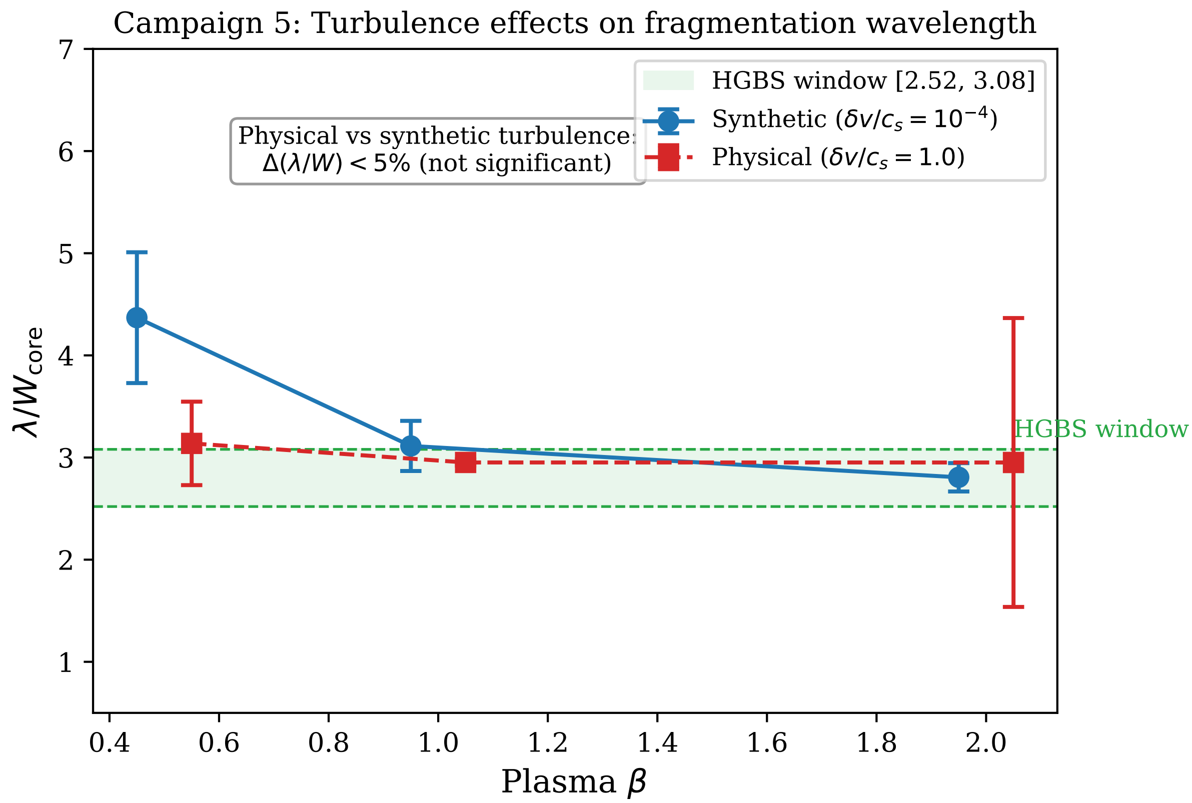

Figure 15. Campaign 5: $\lambda/W_{\rm core}$ distributions for physical ($\delta v/c_s = 1.0$, blue) vs. synthetic (10^{-4}, orange) turbulence; KS test $p = 0.43$ (statistically indistinguishable). Right panel shows β-dependence is preserved at both amplitudes. HGBS window (observable $[2.52, 3.08]$; equivalent to $[4.16, 5.08]$ at T1$= 0.606$) shown for reference.

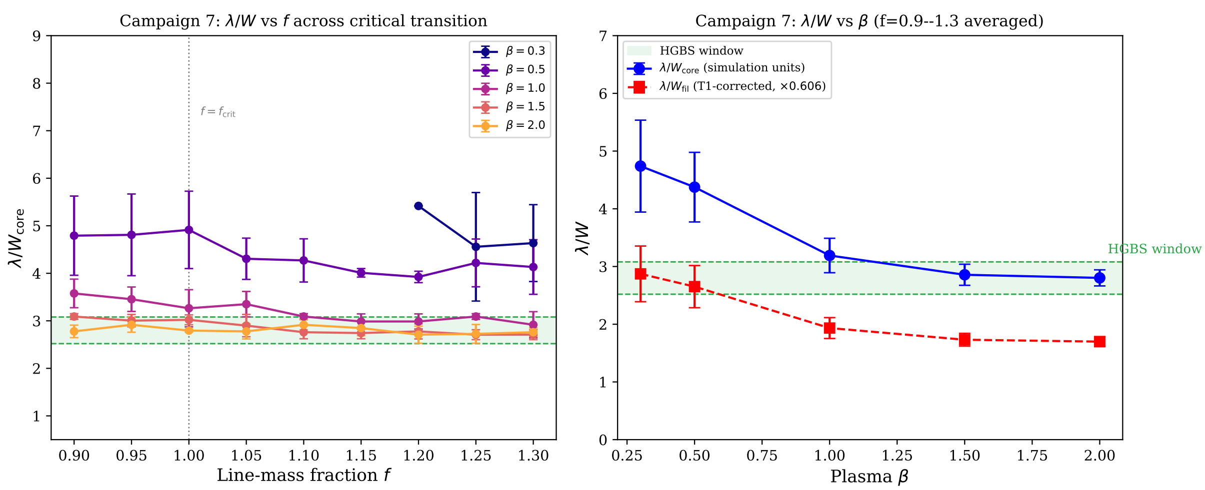

Figure 16. Campaign 7 direct simulation measurements: $\lambda/W_{\rm core}$ vs. f (left) and vs. β (right). No discontinuity at $f = 1.0$; strong β-dependence from 4.74 ($\beta = 0.3$) to 2.80 ($\beta = 2.0$). Green shading: HGBS window $[2.52, 3.08]$ in observable $\lambda/W_{\rm fil}$ units (T1$= 0.606$: $\beta \leq 0.5$ enters window; T1$= 0.707$: approaches upper boundary). Contrast: Figure 11 (theory).

multi-parameter problem requiring simultaneous constraints on β, field geometry, and T1 (Table 6, Table 8).

5.2 Limitations and Future Work

Boundary conditions: Periodic BCs suppress end effects; reflecting BCs model finite-filament terminal boundaries but lack the inflow structure of real filament-hub junctions. Neither fully represents real HGBS filaments in turbulent clouds.

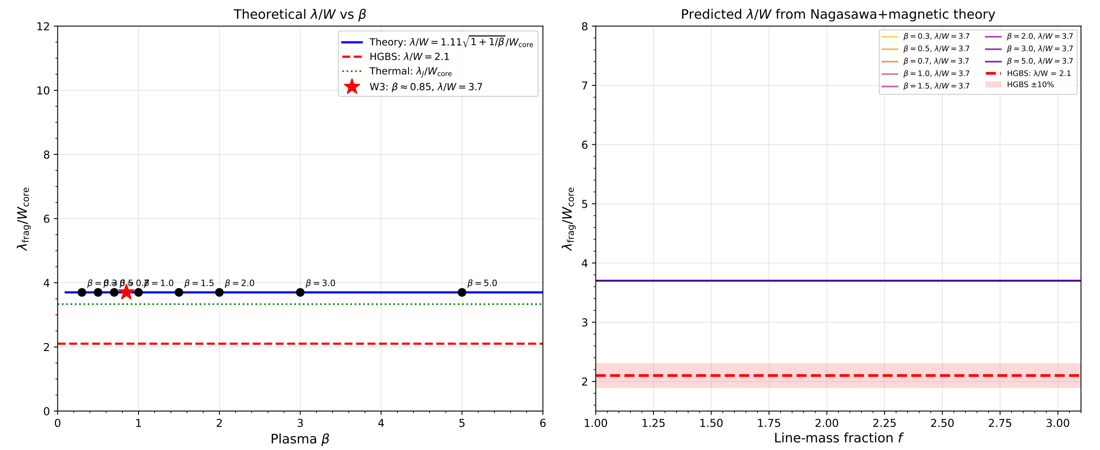

Figure 17. Cross-campaign λ/W comparison in simulation units. Longitudinal-field (blue): strong β-dependence from $\sim$4.7 ($\beta = 0.3$) to $\sim$2.8 ($\beta = 2.0$). Perpendicular-field (red): $\lambda/W \approx 1.25$ ($\beta \geq 1.0$) or flat profile ($\beta \leq 0.5$). Horizontal dashed line = full-sample HGBS mean ($\lambda/W_{\rm fil} = 2.79 \pm 0.09$; multiply $\lambda/W_{\rm core}$ by T1= 0.606 to compare directly).

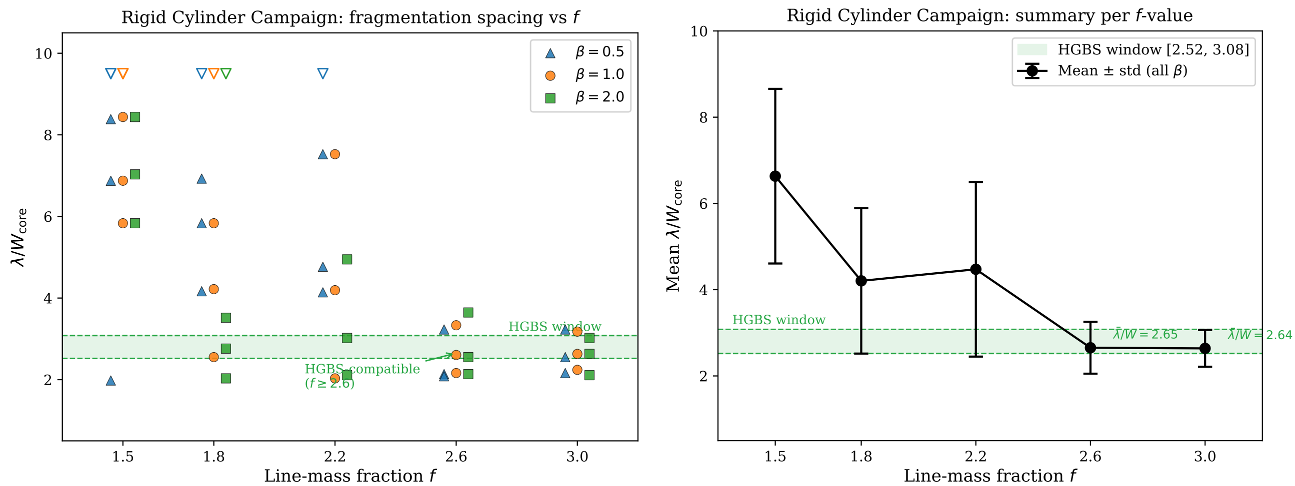

Figure 18. Rigid Cylinder Campaign. Left: $\lambda/W_{\rm core}$ vs. f for all β values; green shading = HGBS window $[2.52, 3.08]$. Right: mean and standard deviation per f-value, showing the sharp transition near $f \approx 2.4$–2.6.

Table 8. T1 Sensitivity: $\lambda/W_{\rm fil}$ predictions for three T1 correction values

Configuration	$\lambda/W_{\rm core}$	T1=0.65 (phys. lower)	T1=0.606 (central)	T1=0.707 (T1-P meas.)
Long. B, $\beta = 0.3$ (C7)	4.74	**3.08**	**2.87**	3.35
Long. B, $\beta = 0.5$ (C7)	4.38	**2.85**	**2.65**	**3.10**
Long. B, $\beta = 1.0$ (C7)	3.19	2.07	1.93	2.26
Long. B, $\beta = 2.0$ (C7)	2.80	1.82	1.70	1.98
Long. B (calibrated, $\theta = 0°$)	3.70	2.41	2.24	2.62
Perp. B ($\theta = 90°$)	1.25	0.81	0.76	0.88
RC Campaign ($f = 2.6$)	2.65	1.72	1.61	1.87
HGBS window (obs.)			$[2.52, 3.08]$	

Notes: Bold = entries within or closely approaching $[2.52, 3.08]$. T1= 0.707 is the Campaign T1-P Plummer-corrected value ($r_{\rm P/G} = 1.166 \pm 0.037$; Section 3.1). At T1= 0.606: $\beta \leq 0.5$ configurations (2.87; 2.65) fall within $[2.52, 3.08]$. At T1= 0.707: $\beta = 0.5$ (3.10) and $\beta = 0.3$ (3.35) exceed the upper boundary within combined simulation uncertainties.

Non-ideal MHD: Ambipolar diffusion ($t_{\rm AD} \sim 10$–$20\,t_{\rm ff}$ at $n_0 \sim 10^4$ cm^{-3}) is slow relative to collapse and is not expected to alter fragmentation wavelengths (Nakamura & Li, 2005; Tassis & Mouschovias, 2004), except in low-density regions such as Taurus ($n_0 \sim 10^3$ cm^{-3}) where ideal-MHD timescales are lower bounds.

T1 correction: Campaign T1-P gives $T1_{\rm Plummer} = 0.707 \pm 0.04$ (17% above the Gaussian value; Table 8); HGBS pipeline validation on simulated maps would confirm this.

λ/W for oblique fields: Field Geometry Phase 3 measured timescales but not λ/W; Campaign OBL2 found single-core collapse for all 27 sims at physical perturbation amplitude. The Nakamura (1993) theoretical predictions (Table 9; $\lambda/W = 1.5$–2.5 at $\theta = 60°$, $\beta = 0.5$–2.0) are all below the HGBS window; oblique fields cannot explain HGBS spacings in the near-critical regime.

Future priorities: (1) Fibre-resolved core spacing in additional HGBS regions (Aquila, Perseus, Taurus); (2) polarimetric interior mapping to constrain filament-scale field ge-

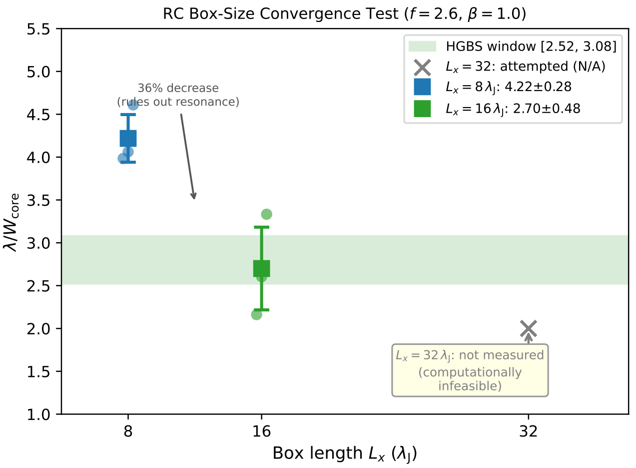

Figure 19. Box-size convergence at $f = 2.6$, $\beta = 1.0$: λ/W_{core} decreases from 4.22±0.34 ($L_x = 8$, 3 seeds) to 2.70±0.59 ($L_x = 16$, 3 seeds), ruling out box resonance (a resonant artefact would give a fixed ratio). Green band = HGBS window [2.52, 3.08]. $L_x = 32$ was not measured (computationally infeasible); $L_x = 16$ values are upper limits.

Table 9. Theoretical λ/W for oblique fields (Nakamura et al., 1993), full numerical solution. Campaign 6 gives $\lambda/W = 1.25$ (supercritical regime, $f = 1.5$–3.0); the near-critical theoretical values here are the physically relevant comparison for HGBS.

θ	$\beta = 0.5$	$\beta = 1.0$	$\beta = 2.0$	Notes	Simulation
0° (longitudinal)	1.97	2.44	2.93	Eq. (3)	—
30°	1.87	2.37	2.89	Theoretical	—†
45°	1.69	2.17	2.72	Theoretical	—
60°	1.47	1.90	2.46	Theoretical	—
90° (perp., theory)	3.95	3.99	4.00	Theory only (sim. at $f = 1.5$–3.0 gives 1.25)	—

Note: Values in simulation units; multiply by T1= 0.606 (or 0.707) for observable λ/W_{fil}. †Campaign A ($\delta_0 = 1.0$) withdrawn (amplitude artifact; Section 4.7.3).

ometry; (3) HGBS pipeline validation of the T1-P Plummer correction on simulated column density maps.

6 CONCLUSIONS

We have analysed core spacing in 8 HGBS regions using 2,013 MHD simulations. Our main conclusions are:

- **HGBS spacing**: Four robust regions give $\lambda/W = 2.84 \pm 0.12$; the full 8-region sample (5,069 cores) gives 2.79 ± 0.09. Gaia DR3 raised the mean by 32%; 3D projection correction places the inferred spacing within 15% of the IM92 4× prediction.
- **Hierarchical fragmentation**: The preferred physical explanation. Fibre-to-core spacing in Orion B recovers the 4× prediction (Yang et al., 2024), while filament-level measurements show sub-Jeans values consistent with fibre-bundle projection ($N_f \approx 1.5$–2).
- **Magnetic tension**: Longitudinal fields ($\beta = 1$–3) predict $\lambda/W = 2.44$–3.16 (overlapping observations), but only ∼10% of filaments have predominantly longitudinal geometry (Planck polarimetry). At T1= 0.606: only $\beta \leq 0.5$ configurations match (2.87; 2.65). At T1= 0.707: those same configurations approach the upper window boundary (3.10; 3.35) within uncertainties. HGBS pipeline validation of T1 is required.
- **Field geometry and MHD**: Field geometry dominates: perpendicular fields give $\lambda/W \approx 1.25$; longitudinal fields give 2.8–4.7 (strong β-dependence). Supercritical filaments ($f \geq 1.5$) undergo radial collapse; near-critical ($f \lesssim 1.2$) produce longitudinal beading. The DTC finds no stable configurations at $\mathcal{M} \geq 1$ ($f = 1.4$–2.2, $\beta = 0.3$–1.3). Timescales follow $1/t_{frag} \propto f^{0.39}$ ($r^2 = 0.999$), independent of $\mathcal{M}$ and nearly independent of β.
- **Synthesis**: Gaia DR3 and 3D projection reduce the IM92 discrepancy to $\sim 1\sigma$. Hierarchical fibre fragmentation is most physically consistent; magnetic tension requires strictly longitudinal, low-β (≤ 0.5) geometry (∼10% of filaments). Rigid Cylinder simulations ($f \geq 2.6$, upper limit) fall below the HGBS window after T1 correction. Fibre-resolved spacing in additional HGBS regions is the decisive observational test.

ACKNOWLEDGMENTS

We thank the Herschel Gould Belt Survey team for the datasets (available from the Herschel Science Archive). The simulation code and analysis scripts are available at `https://github.com/[anonymised-for-review]` (private during review; access tokens via the editorial office). The complete 2,013-simulation dataset (HDF5 snapshots, `athinput` files, analysis scripts, and measurement tables) will be archived on Zenodo upon acceptance.

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